

HW 05

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Exercise 1

Collaborators: Sam

Tools and resources used: I used a lot of stack exchange and a little Gemini.

Exercise 2

- Points attempted: 3
- Self-assessed score: 3.
- Time spent: 3 minutes.
- Difficulty: 1/10.
- Questions/Feedback: None.

Done: For the blog post, I am looking to do something related to the topology prompt. I am currently only in a group of 2, and that is with Justin.

Exercise 3

- Points attempted: 10
- Self-assessed score: 10.
- Time spent: 60 minutes.
- Difficulty: 6/10.
- Questions/Feedback: Correctness, clarity.

Problem 7. 7.1 If G is a group of infinite order, then all subgroups of G have infinite order.

False: A counterexample would be that the group of $\{1, -1\}$ under multiplication is a subgroup of the group of non-zero rationals under multiplication.

7.2 If G is an abelian group, then all subgroups of G are abelian groups.

True: Commutativity applies to all elements of G , so by definition a subgroup of the elements of G must also be commutative.

7.3 If H is a normal subgroup of $(G, *)$, then $g * h = h * g$ for all $g \in G$ and $h \in H$.

False: The normal subgroup invariant only requires that the cosets $gH = Hg$, however, it doesn't necessarily mean that $g * h = h * g$, just that $g * h$ and $h * g$ are in the same coset $gH = Hg$.

7.4 Suppose that $(X, \star)$ is a set with a binary operation that is not necessarily a group, and there exists a "left identity" $x \in X$ such that $x \star a = a$ for all $a \in X$. Then x is also a "right identity", that is $a \star x = a$ for all $a \in X$.

False: Imagine that the binary operation $\star$ just selects the element on its right, so all elements are the "left identity" and no elements are the "right identity."

7.5 If G is a group and $g \in G$, then every column of the Cayley table for G contains g exactly once.

True: we know this to be true because for every $g' \in G$, $g * g^{-1} * g' = g'$, so that must be a column of g .

7.6 The set of rational numbers $\mathbb{Q}$ forms a group under multiplication.

False: There is no inverse of 0.

7.7 If G is a finite group of order n , then for every element $g \in G$, $g^n = e$.

True: the order m of all subgroups H must be a factor of G , so considering all subgroups generated by a single element $g \in G$, the order of $\langle g \rangle$ must be a factor of the order of G , so $g^n = e^{n/m} = e$.

Problem 8 Let G be a (not necessarily abelian) group, and let H be a (not necessarily) subgroup of G . Let the normalizer of H be defined as

$$N_G(H) = \{g \in G \mid gH = Hg\}.$$

8.1 Prove that $N_G(H) \leq G$.

Proof. We begin by noticing that $N_G(H)$ is a subset of G . Thus, the group axiom of associativity, is already satisfied.

First we show that $N_G(H)$ has an identity element. Notice $e_G \in N_G(H)$ because $e_G H = H e_G$. So, e_G is the identity element of $N_G(H)$.

Next, assuming that $n \in N_G(H)$, we can show that n^{-1} must also be in $N_G(H)$. First, we see that for every n and $h_1 \in H$, there exists a unique $h_2 \in H$, such that $n * h_1 = h_2 * n$. Now, we left and right multiply by n^{-1} , to get

$$n * h_1 = h_2 * n \tag{1}$$

$$n^{-1} * n * h * n^{-1} = n^{-1} * h_2 * n * n^{-1} \tag{2}$$

$$h * n^{-1} = n^{-1} * h_2. \tag{3}$$

Therefore, $n^{-1} \in N_G(H)$, so every element has an inverse.

Lastly, we show that $N_G(H)$ is closed. That is, for all $n_1, n_2 \in N_G(H)$, $n_1 * n_2 \in N_G(H)$ and $(n_1 * n_2)H = H(n_1 * n_2)$. Let us pick some $h_1, h'_1, h_2, h'_2 \in H$, such that

$$n_1 * h_1 = h'_1 * n_1 \tag{4}$$

$$n_2 * h_2 = h'_2 * n_2. \tag{5}$$

We isolate n_1 in equation (4) as $n_1 = h'_1 * n_1 * h_1^{-1}$. We left multiply equation (5) by n_1 to get

$$n_1 * n_2 * h_2 = h'_1 * n_1 * h_1^{-1} * h'_2 * n_2. \tag{6}$$

Now, because $n_1 H = H n_1$, there must exist some $h_1'^{-1}$ where $n_1 * h_1^{-1} = h_1'^{-1} * n_1$. We substitute this into equation (6) to get

$$= h'_1 * h_1'^{-1} * n_1 * h'_2 * n_2. \tag{7}$$

Similarly for $n_1 * h'_2$, there exists some h_2'' where $n_1 * h'_2 = h_2'' * n_1$. So, we have,

$$= h'_1 * h_1'^{-1} * h_2'' * n_1 * n_2. \tag{8}$$

Since H itself is closed, $h'_1 * h_1'^{-1} * h_2'' \in H$. Therefore, $n_1 * n_2$ is also a normalizer $N_G(H)$. Thus, $N_G(H)$ is closed, so $N_G(H)$ is a subgroup of G . $\square$

8.2 Prove that for all $g \in N_G(H)$ and $h \in H$, $ghg^{-1} \in H$.

Proof. We begin by noticing that $gH = Hg$. That is, there for every $h \in H$, there exists a unique $h' \in H$ where $g * h = h' * g$. We right multiply by g^{-1} to get $g * h * g^{-1} = h'$. Thus, for all $g \in N_G(H)$ and $h \in H$, $ghg^{-1} \in H$. $\square$